

Lab 1-1

1. Please briefly describe the idea of the bubble sorting algorithm, implement it through programming, and count the execution numbers of the comparison steps in the algorithm when the input scale n is 10, 100, 1000. The submitted content includes the idea of the bubble sorting algorithm, program code, and screenshots showing the output of the program. (Input array randomly generated)
2. Please briefly describe the idea of the insertion sorting algorithm, implement it through programming, and count the execution numbers of the comparison steps in the algorithm when the input scale n is 10, 100, 1000. The submitted content includes the idea of insertion sorting algorithm, program code, and screenshots showing the output of the program. (Input array randomly generated)
3. Please briefly describe the idea of the merge sorting algorithm, implement it through programming, and count the execution numbers of the comparison steps in the algorithm when the input scale n is 10, 100, 1000. The submitted content includes the idea of merge sorting algorithm, program code, and screenshots showing the output of the program. (Input array randomly generated)

The output example is as follows:

```
n=10
The array before sorting:217 1743 666 1317 1581 1692 1610 1068 1526 489
The sorted array:217 489 666 1068 1317 1526 1581 1610 1692 1743
Number of algorithm comparisons:45
```

```
n=100
Number of algorithm comparisons:4950
```

```
n=1000
Number of algorithm comparisons:499500
```

